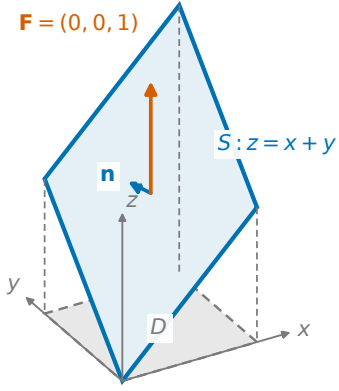
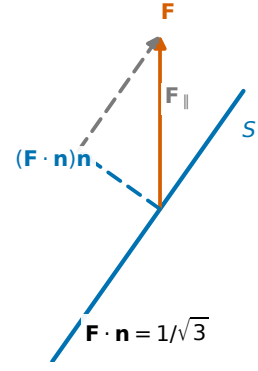
A



$\text{Area}(S) = \sqrt{3}, \quad \text{Area}(D) = 1$

B



$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS = (1/\sqrt{3})\sqrt{3} = 1$